Supporting Information for

Thermal-Conductivity-Driven Dual-Wedge Thermoelectric Architecture for Intelligent Thermal Perception and Material Identification

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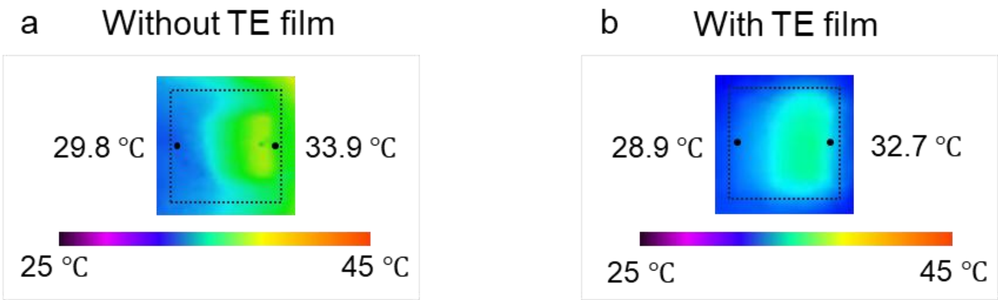

Figure S1. Top-view infrared thermal images of the DWTE architecture (a) without (b) and with thermoelectric layer under the same vertically applied thermal stimulus.

As shown in Figure S1, the temperature distributions and the resulting temperature differences are nearly identical in the presence and absence of the thermoelectric layer. It indicates that the contribution of thermal diffusion of the thermoelectric material layer to the temperature difference and thermoelectric performance of DWTE device is minimal. We attribute this behavior to the ultrathin nature of the thermoelectric film, for which the associated thermal diffusion is insufficient to noticeably perturb the temperature field primarily governed by the dual-wedge architecture.

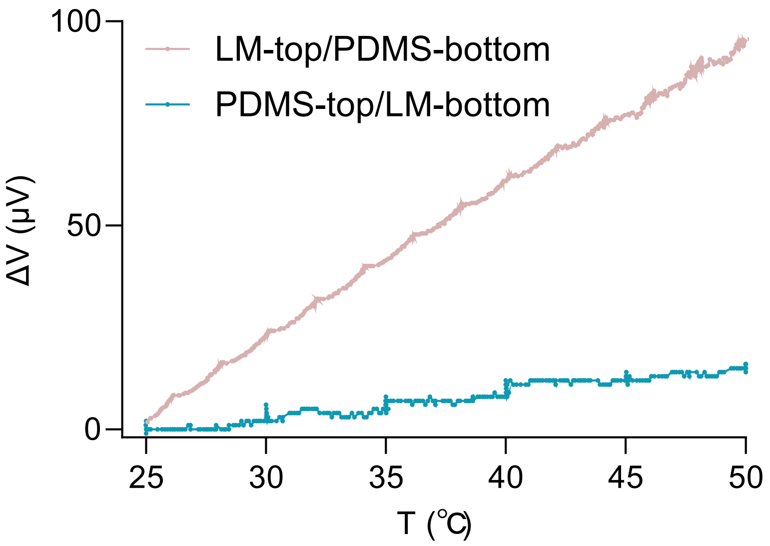

Figure S2. Experimental thermoelectric output voltage (ΔV) for DWTE devices with different stacking sequences (LM-top/PDMS-bottom and PDMS-top/LM-bottom). The cold side was fixed at 25 °C and the hot side was gradually increased to 50 °C.

Under identical thermal conditions (cold side fixed at 25 °C and hot side increased to 50 °C), the original LM-top/PDMS-bottom configuration exhibits a significantly higher thermoelectric output, reaching ~95 μV at 50 °C, whereas the reversed stacking produces only ~15 μV (Figure S1). This result clearly demonstrates that the stacking sequence plays a critical role in establishing an effective in-plane temperature difference.

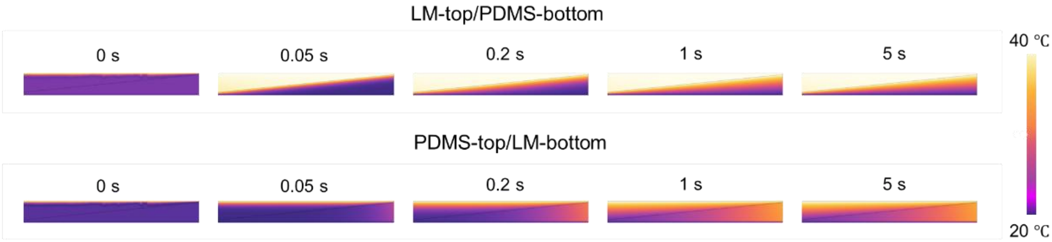

Figure S3. Finite-element simulations for the DWTE structures with LM-top/PDMS-bottom and PDMS-top/LM-bottom under the same thermal boundary condition.

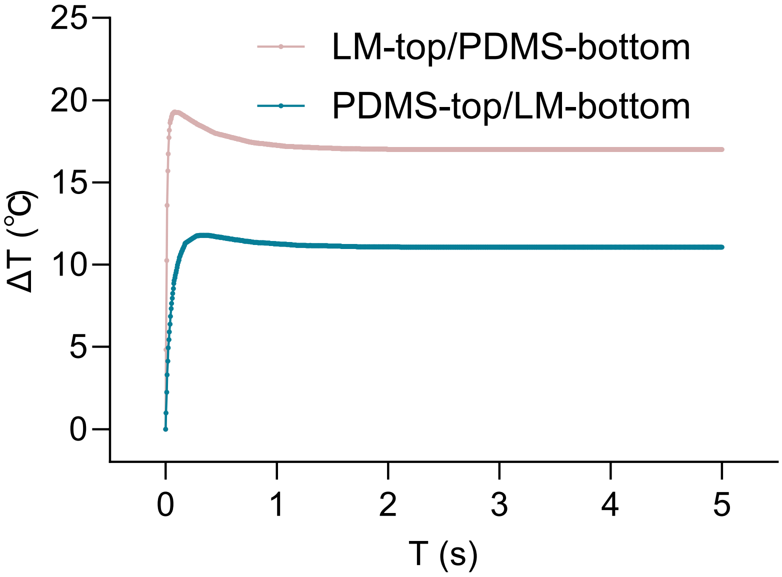

Figure S4. Simulated evolution of the effective temperature difference (ΔT) relevant to thermoelectric output for the two stacking sequences under the same thermal stimulus (ΔT=20 °C).

Finite element simulations further elucidate the underlying mechanism. In the LM-top/PDMS-bottom configuration, the high-thermal-conductivity LM near the hot side facilitates rapid heat delivery, while the PDMS layer suppresses vertical heat diffusion, enabling the formation of a strong and self-sustained lateral temperature gradient. In contrast, placing PDMS at the top layer significantly increases the thermal resistance and hinders heat transport toward the thermoelectric layer, thereby weakening the its thermal response and suppressing the generation of a lateral temperature difference (Figures S2 and S3).

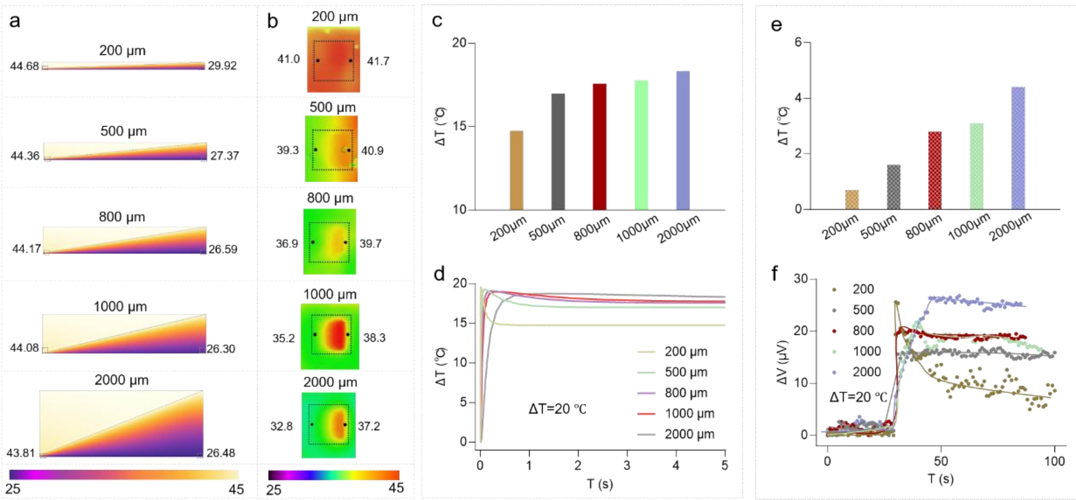

Figure S5. Thickness-mediated thermal gradient and thermoelectric response under steady-state conditions. (a) Finite element simulations of steady-state heat transfer in DWTE devices with various dual-wedge thicknesses (200, 500, 800, 1000, and 2000 μm) under a 20 K thermal stimulus. **(b)** Top-view infrared thermal images showing the lateral temperature distribution across DWTE structures with different thicknesses under steady-state conditions. **(c)** Simulated steady-state ΔT values for DWTE devices with varying thicknesses under a 20 K stimulus. **(d)** Variation in lateral ΔT between the hot and cold sides of DWTE-based devices obtained from finite element simulations under steady-state conditions. **(e)** Experimentally measured ΔT value between the hot and cold sides under a 20 K stimulus, captured by infrared thermal imaging under steady-state conditions. **(f)** Measured time-dependent thermoelectric output voltages of DWTE devices with different thicknesses under a 20 K steady-state stimulus.

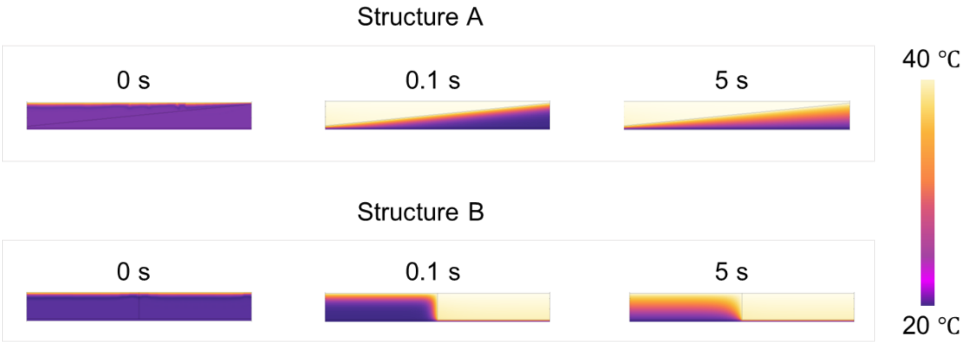

Figure S6. Time-resolved finite-element temperature maps comparing heat-transfer evolution in Structure A and Structure B under the same applied thermal stimulus.

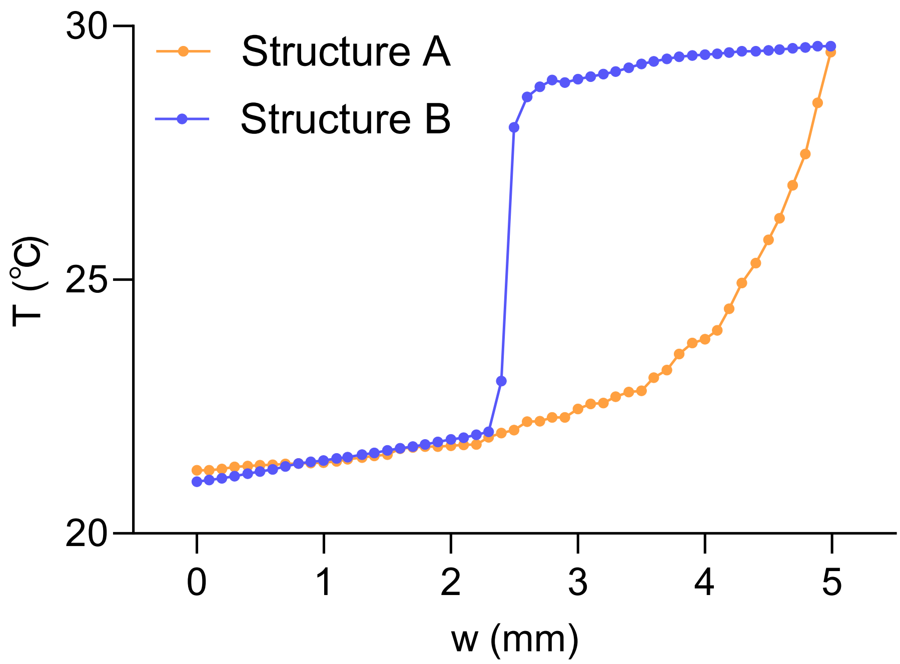

Figure S7. Simulated surface temperature profile along the thermoelectric film length for Structure A and Structure B under the same applied thermal stimulus.

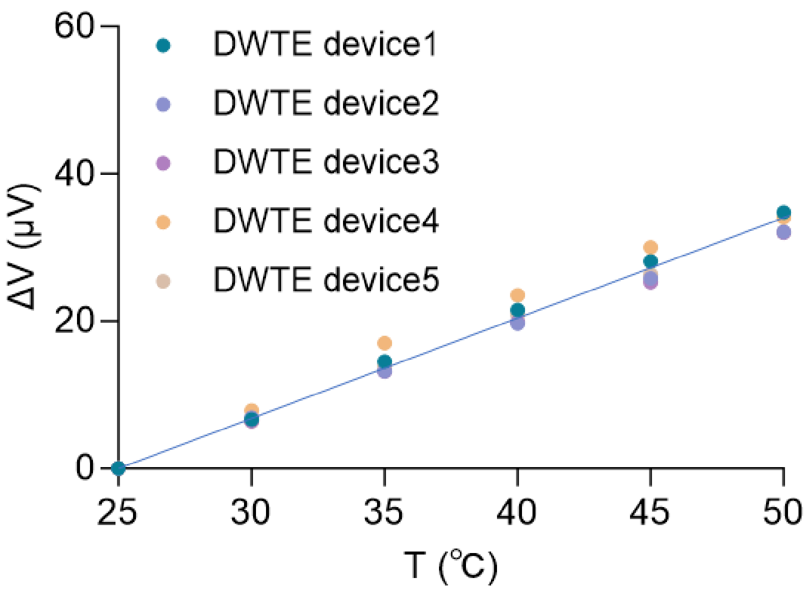

Figure S8. Batch-to-batch fabrication consistency of DWTE devices.

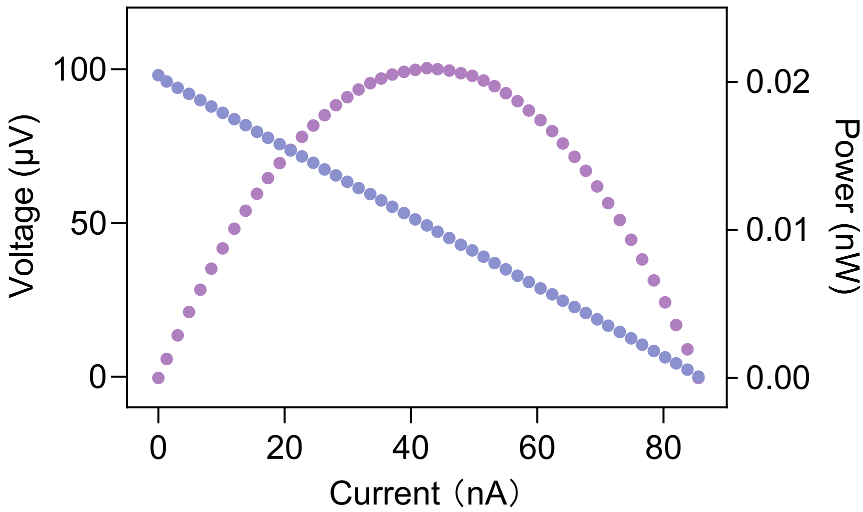

Figure S9. Output voltage and output power versus current density curves of the DWTE device, indicating potential applications for energy conversation.

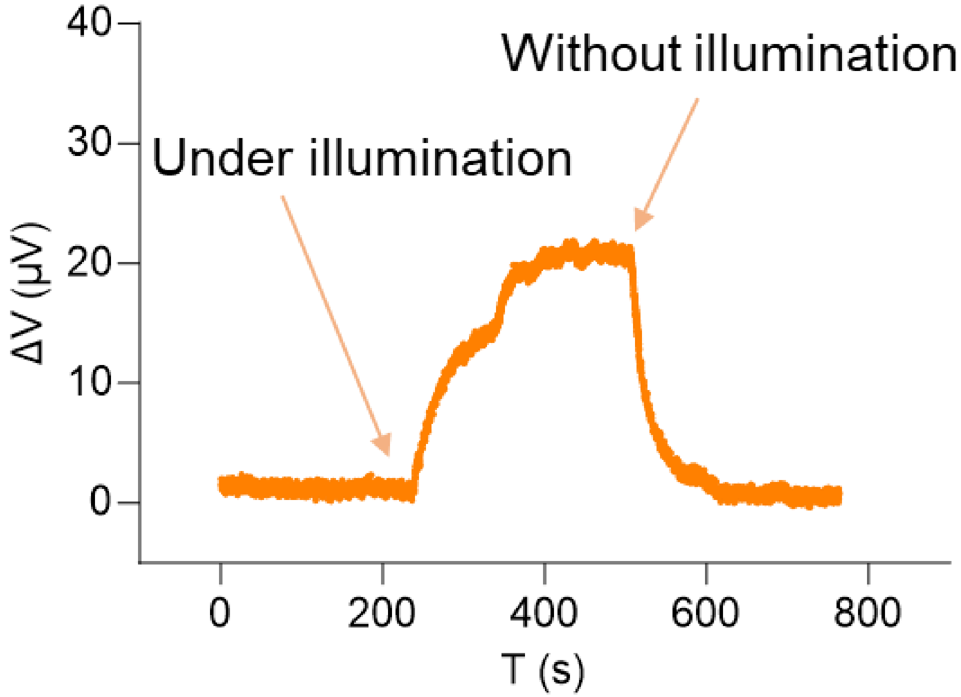

Figure S10. Photothermal conversion performance of the DWTE device under xenon-lamp illumination.

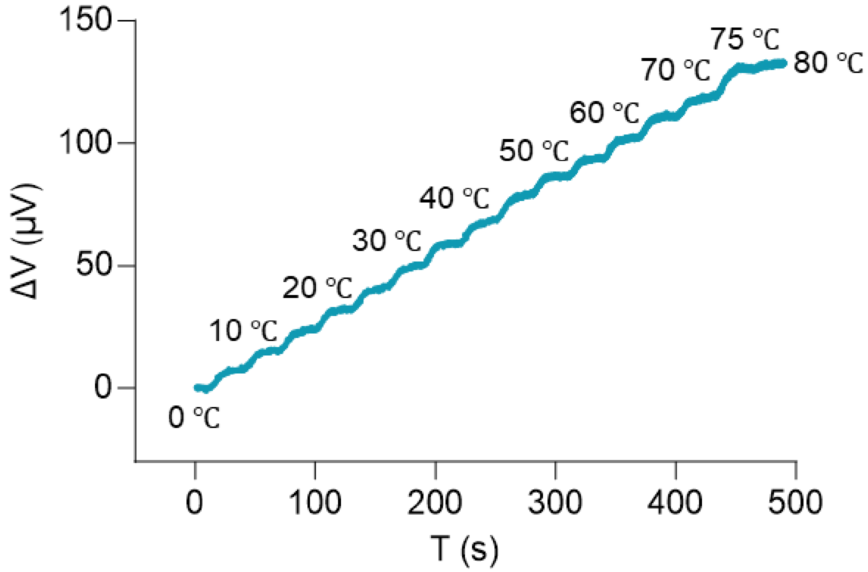

Figure S11. Temperature detection range of the DWTE device.

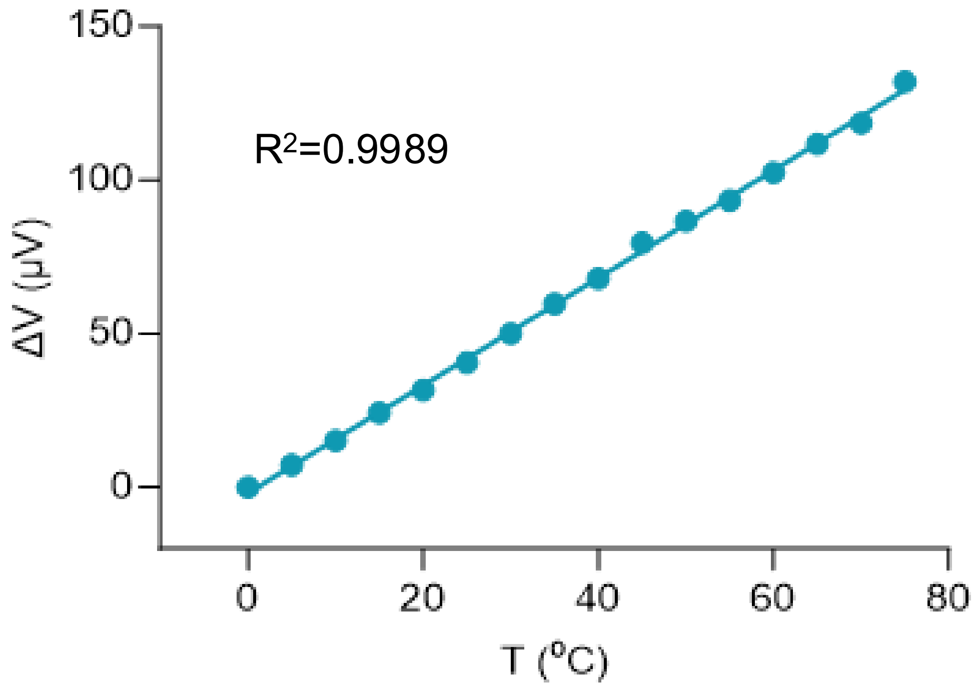

Figure S12. Linearity of the DWTE device.

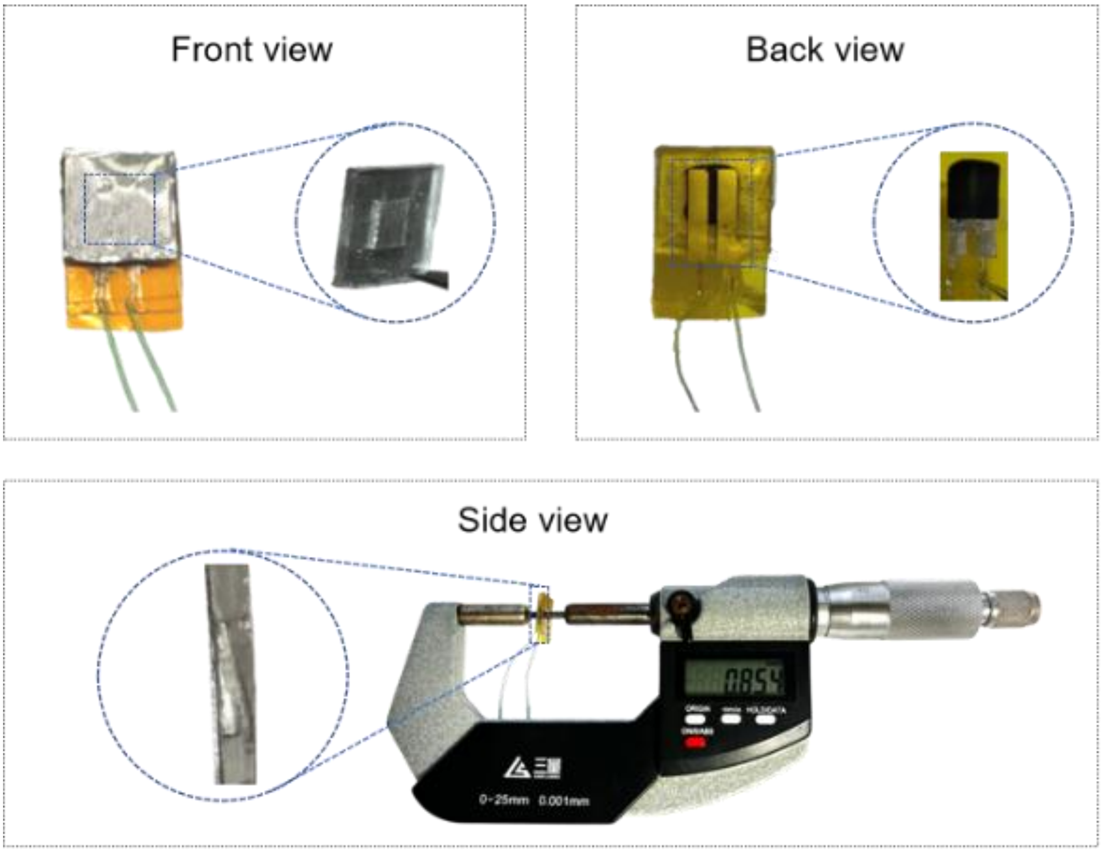

Figure S13. Photographs of the fabricated DWTE device from multiple perspectives.

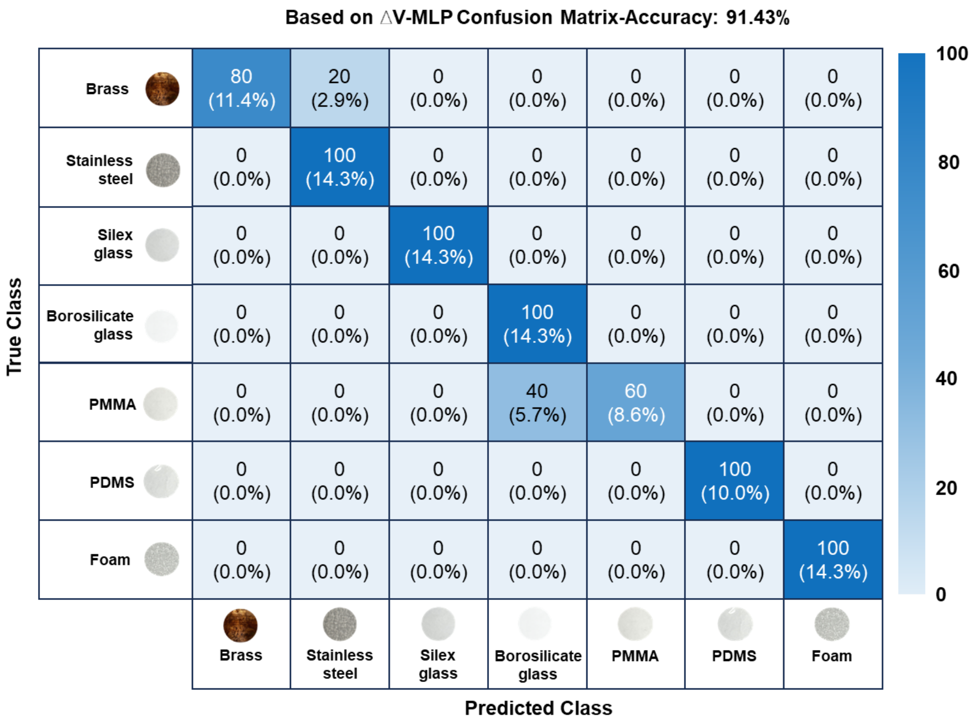

Figure S14. Confusion matrix for material classification using ΔV as the input feature.

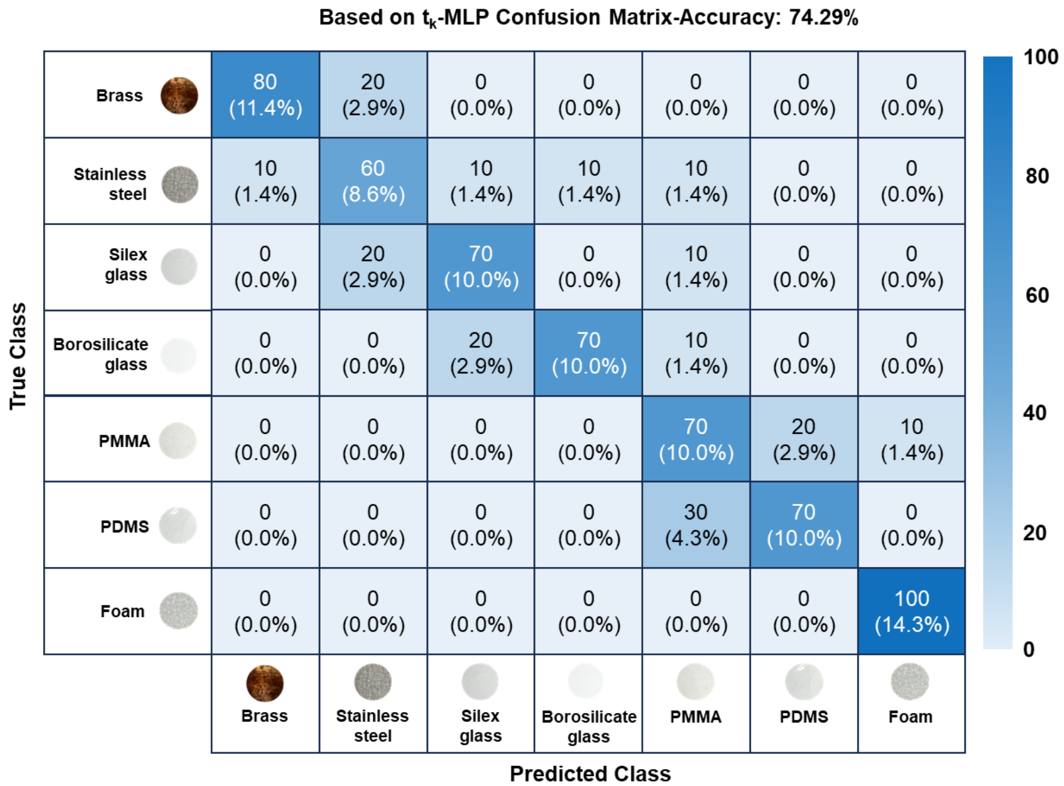

Figure S15. Confusion matrix for material classification using t_k as the input feature.

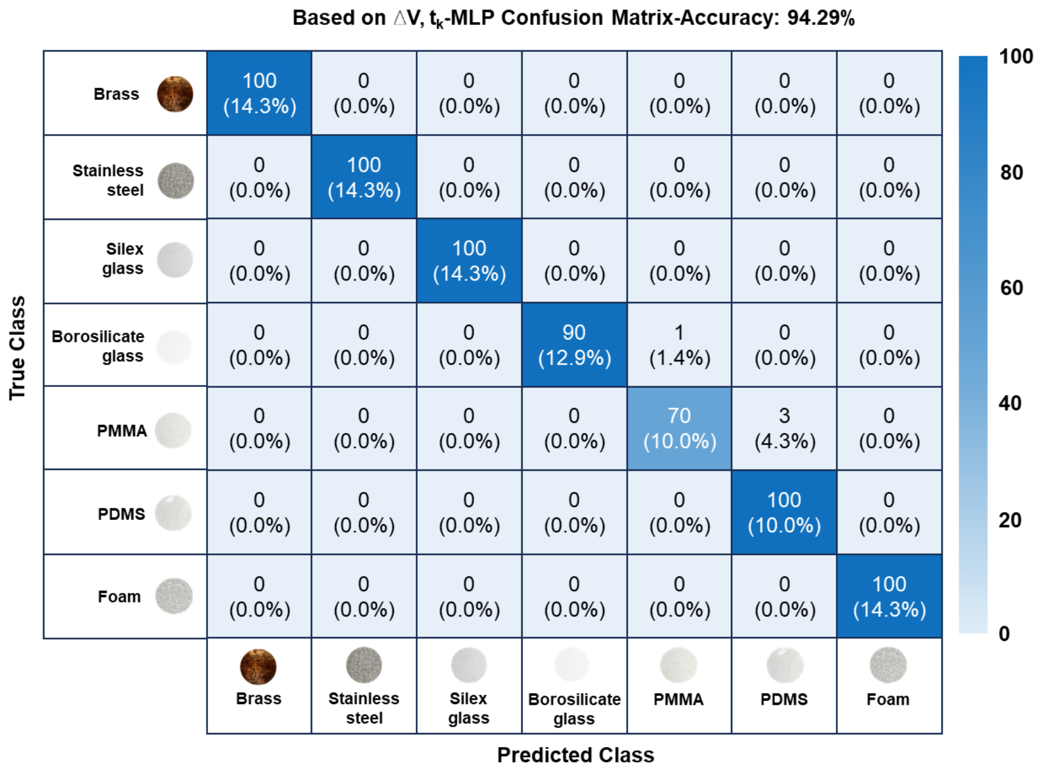

Figure S16. Confusion matrix for material classification using the 2D feature vector (ΔV, t_k) as the input feature.

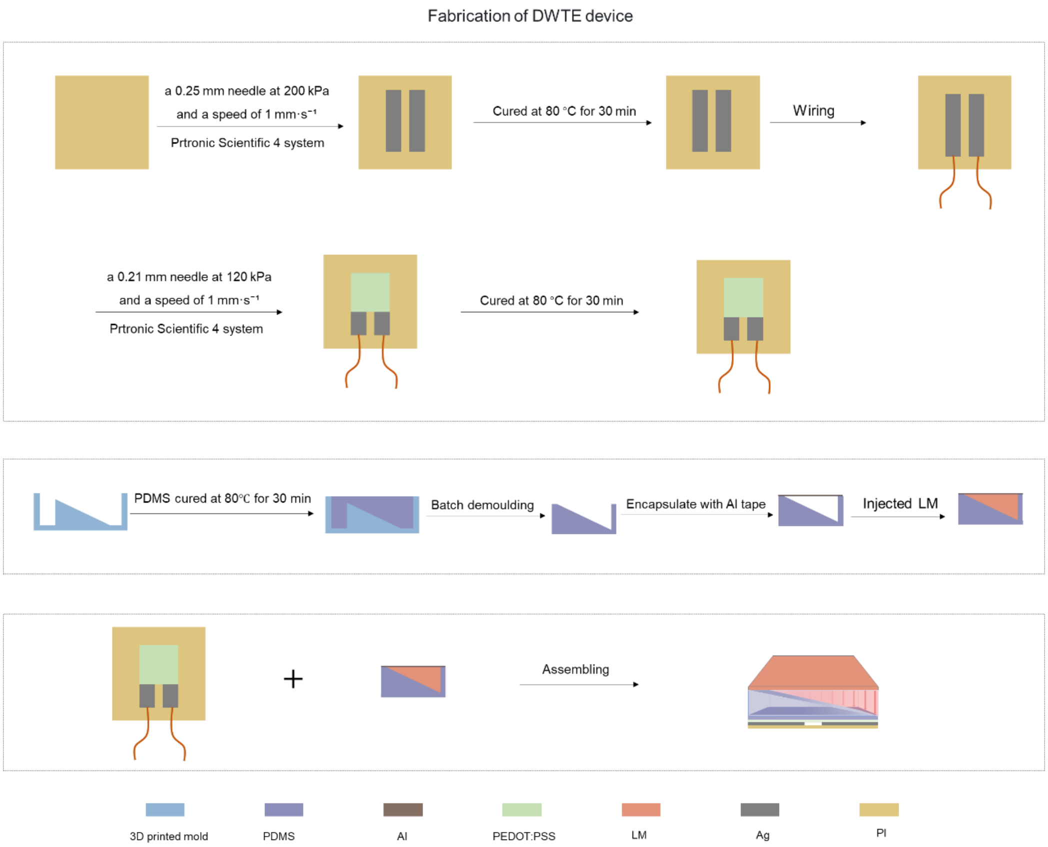

Figure S17. Schematic illustration of the fabrication process for the DWTE device.

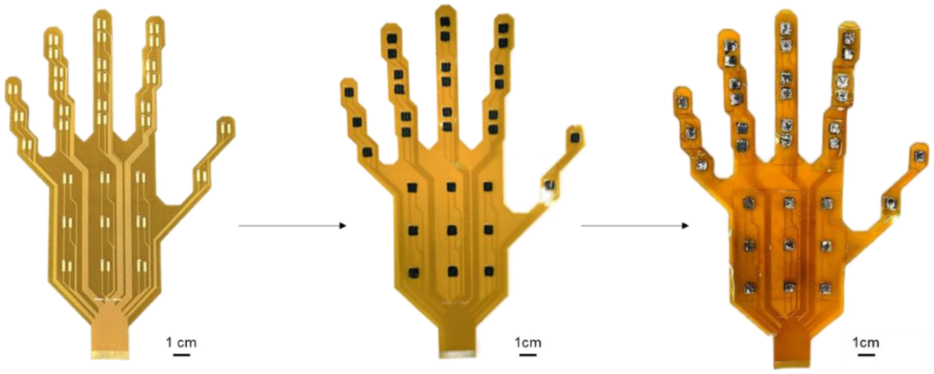

Figure S18. Preparation of the 32-channel DWTE-based thermoelectric sensor array.

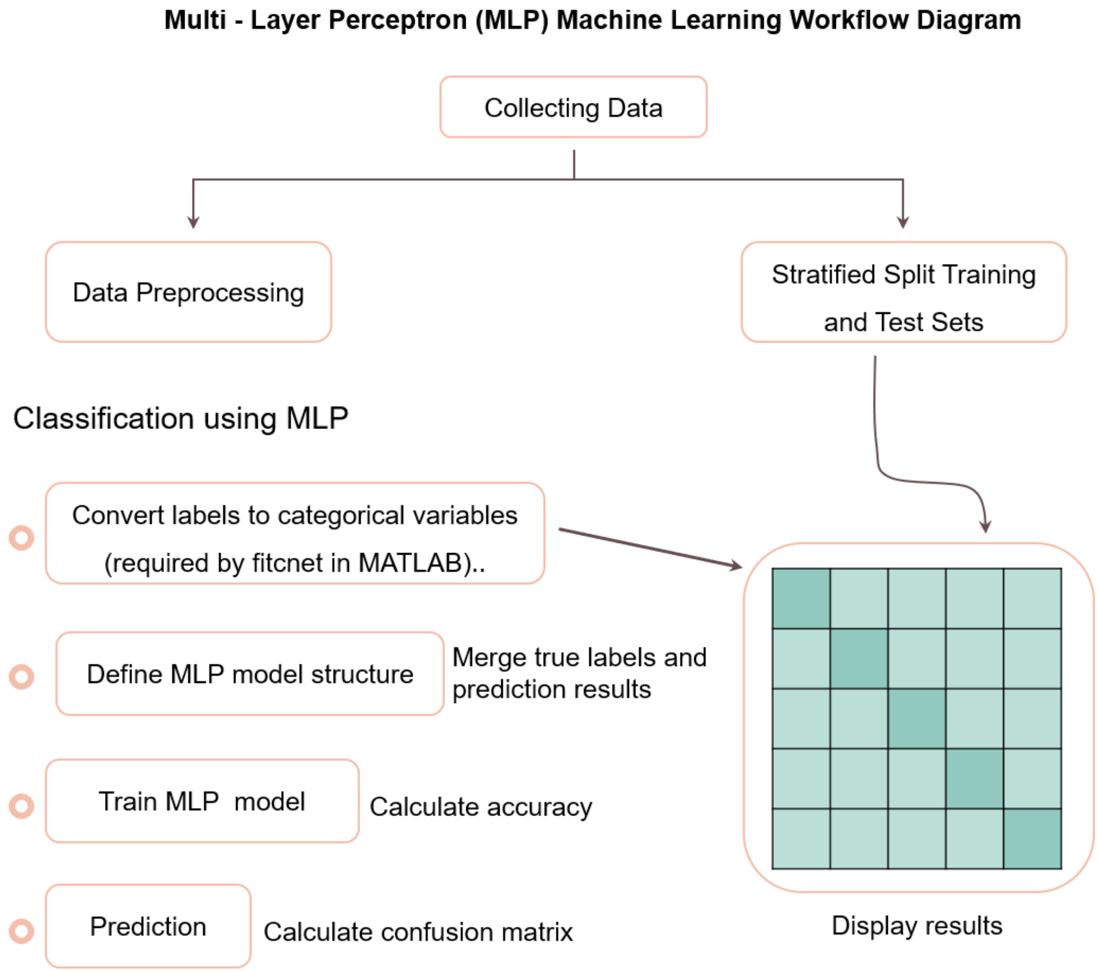

Figure S19. Multilayer Perceptron (MLP) neural network workflow

A stratified 60:40 train-test split was applied to preserve class balance across subsets, and target labels were converted to categorical variables, as required by the fitcnet function. The MLP model architecture was then defined by specifying the number of hidden layers, neuron counts, and activation functions. Model training was conducted on the training set using appropriate hyperparameters. The trained model was subsequently used to predict class labels for the test set, and performance was evaluated by comparing predicted outputs with ground-truth labels. A confusion matrix was generated to visualize class-wise prediction outcomes, and the overall identification accuracy was calculated as the percentage of correctly classified samples in the test set.

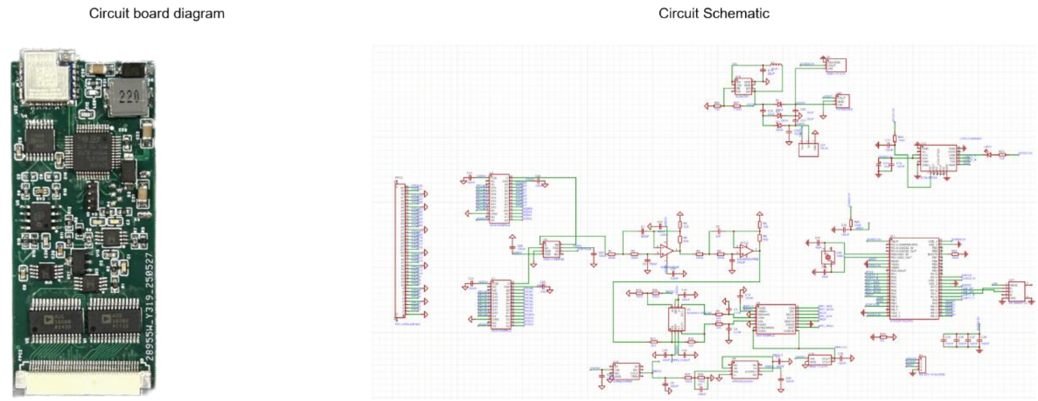

Figure S20. Circuit board and circuit design diagram of a 32-channel signal acquisition system

Analog thermoelectric signals (in the microvolt range) from all 32 channels were sequentially sampled through a two-stage multiplexing circuit. In the first stage, two parallel 16-to-1 analog multiplexers selected groups of input channels; in the second stage, a 2-to-1 multiplexer completed final channel selection, enabling time-division sampling of all 32 inputs. The selected analog signal was amplified 100-fold using a precision amplifier to meet the input requirements of subsequent circuits. The amplified signal was then passed through a low-pass filter (cutoff frequency: 10 Hz) to suppress high-frequency noise, and subsequently digitized via an analog-to-digital converter (ADC). A microcontroller unit (MCU) coordinated the sampling sequence and issued control commands to the ADC. The resulting digital signals were transmitted wirelessly to a custom-developed software platform, which performed real-time signal parsing, visualization, and data storage.

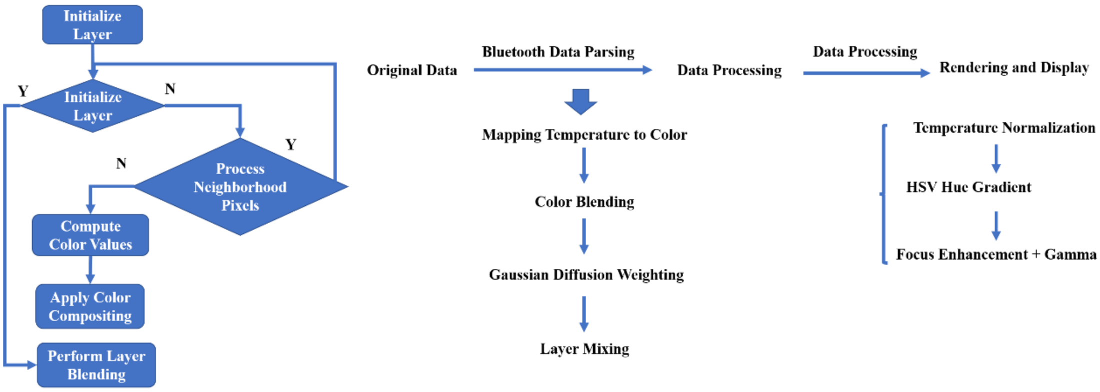

Figure S21. Thermal Visualization Methodology for Palmar Temperature Distribution

To achieve intuitive and precise visualization of temperature distribution in the palm region, 32 sets of data $\{X_i, V_i\}$ obtained by parsing Bluetooth-transmitted raw data are used to construct a palm thermal map based on the palm layer. The overall mechanism includes data normalization and color mapping, Gaussian diffusion modeling, focused temperature region enhancement, and perception regulation mechanisms to improve the overall display effect.

I. Data Normalization and Color Mapping Mechanism

The first step in thermal map rendering is to map the temperature value V_i of each data measurement point to a standardized scale r ∈ [0,1], calculated as follows:

$$r = \frac{V_i - V_{min}}{V_{max} - V_{min}}$$

Based on the normalized data, HSV (Hue–Saturation–Value) hue interpolation is adopted. Temperature values are interpolated in a linear hue angle, transitioning from blue through green to red, enabling a more natural and continuous color transition.

II. Gaussian Diffusion Modeling

Due to the thermal conductivity of real touch temperatures, each temperature point not only affects its own position but also the surrounding adjacent areas. To simulate this spatial diffusion process, the system uses a two-dimensional Gaussian kernel for modeling:

$$\omega(d) = exp\,(-\frac{d^2}{2\sigma^2})$$

where d is the distance between the pixel and the temperature point, i.e., the distance between X_i and the layer position P_j, and σ controls the diffusion range. In this work,

σ is a fixed parameter that serves as a spatial smoothing scale matched to the sensor distribution and palm dimensions, rather than a physical temperature-dependent parameter that varies with the thermal environment. After a pixel p_j is affected by multiple temperature points, its final color C_j is the weighted average of these colors, with the weight being the output of the Gaussian function. Color superposition adopts a linear mixing method:

$$C_{new} = (1 - \alpha) \cdot C_{prev} + \alpha C_{point}$$

where $\alpha = \omega(d) \cdot$scaling factor, C_{prev} is the existing color of the current pixel, and C_{new} is the mixed color of the current pixel. After the superposition of P_j pixel points, the thermal map layer is overlaid on the palm-shaped mask, displaying color information only in the effective area.

III. Focused Temperature Region Enhancement and Perception Regulation Mechanism

To improve the ability to identify key temperature intervals in the thermal map, a focused temperature region enhancement mechanism (focus stretch) and Gamma correction mechanism are introduced in the color mapping process. This mechanism automatically identifies the most densely distributed temperature range through statistical analysis of all temperature data and sets it as the focus region of color mapping, thereby enhancing the color resolution of this interval and highlighting its physiological change characteristics.

Focus stretching nonlinearly remaps the normalized ratio r, increasing the proportion of the specified temperature region in color mapping to improve color resolution within this interval; Gamma correction adjusts the nonlinearity of the color response curve, further enhancing the visual sensitivity to temperature gradients. The combination of the two can effectively amplify the color contrast in key temperature regions and highlight physiological change characteristics.

The implementation method is as follows: from each frame of data $\{T_1, T_2, T_3, ..., T_n\}$, the system first constructs a temperature histogram with a fixed width δ and counts the occurrence frequency of each temperature segment. Subsequently, several temperature ranges with the highest frequencies are selected as candidate high-density temperature regions, and their coverage ranges are extracted as the boundaries of the focused temperature region. Let the selected temperature interval be [T_{low}, T_{high}], then it can be automatically calculated:

$$focusCenter = \frac{T_{low} + T_{high}}{2}, \quad focusWidth = T_{high} - T_{low} + \delta$$

In the color mapping stage, the focused temperature region will occupy a larger color space through nonlinear stretching, thereby enhancing the visual sensitivity to temperature differences in this interval.

IV. Inter-Frame Temperature Interpolation Algorithm and Transition Frame Construction Mechanism

To achieve smooth transition and visual continuity of thermal map animation, a temperature interpolation transition mechanism (inter-frame temperature interpolation) is introduced between the main frames sent by the lower computer (hardware). This mechanism supports automatic generation of several transition frames between any two consecutive main frames, forming a visually continuous and gradient thermal map animation.

Let the two main frames be F_a and F_b, and the system will insert n transition frames (e.g., n=10) between them. The temperature interpolation ratio of the s-th transition frame is:

$$a_s = \frac{S}{n+1} \quad s = 1,2,3 \ldots n$$

Then, the interpolated temperature at each point (X_i, V_i) is:

$$V_k^{(s)} = (1 - \alpha_s) \cdot T_k^{(a)} + \alpha_s V_k^{(b)}$$

The interpolation process keeps the coordinates unchanged and only performs linear transition on the temperature values, finally obtaining a series of continuous frames, thus realizing smooth transition between data transmissions of the lower computer.

Supplementary Note S1.

The baseline open-circuit thermoelectric voltage V(t) was recorded at a constant temperature (no thermal stimulus) with a sampling rate of 10 Hz. The electrical noise was quantified as the RMS (standard deviation) of the baseline voltage fluctuations[1–3]

$$\mu_{base} = \frac{1}{N}\sum_{i=1}^{N} V_i$$

$$\sigma_{\text{base}} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(V_i - \mu_{base})^2}$$

The signal-to-noise ratio (SNR) was then evaluated using the peak response under a small temperature step:

$$SNR_{peak} = \frac{V_{peak} - \mu_{\text{base}}}{\sigma_{\text{base}}}$$

Based on the measured data, the baseline statistics are μ_{base}=0.061 μV and σ_{base}=0.023 μV, and the peak voltage under a ΔT=0.1 ºC step is V_{peak}=0.197 μV, yielding a SNR_{peak} of 5.84. These results confirm the low electrical noise level and the stable output signal of the DWTE device.

Reference

[1] J.-B. Ahn, B. Yoo, D. J. Pines, C.-Y. Kuo, M. Wang, N. S. G. Bejjanki, S. Kim, *Commun Mater* **2024**, *5*, 136.

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